

Unit 3.6 Derivatives of logarithmic functions.

Note Title

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If $y = \log_a x$ determine $\frac{dy}{dx}$

$$y = \log_a x \iff a^y = x$$

$$\Rightarrow a^y \cdot \ln a \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{a^y \ln a} = \frac{1}{x \ln a}$$

$$\text{Therefore } \frac{d}{dx} \ln x = \frac{1}{x \ln e} = \frac{1}{x}$$

$$\frac{d}{dx} \log_a x = \frac{1}{x \ln a}$$

$$\frac{d}{dx} \log_a \underline{f(x)} = \frac{\underline{f'(x)}}{\underline{f(x)} \cdot \ln a}$$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

$$\frac{d}{dx} \ln f(x) = \frac{f'(x)}{f(x)}$$

$$\frac{d}{dx} \ln |x| = \frac{1}{x} \quad \left[\text{See textbook} \right].$$

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Examples:

$$\textcircled{1} \quad y = \cot(\ln x) \\ y' = -\operatorname{cosec}^2(\ln x) \cdot \frac{1}{x}$$

$$\textcircled{2} \quad y = (\ln 2x)^3 \\ y' = 3(\ln 2x)^2 \cdot \frac{2}{2x}$$

$$\textcircled{3} \quad y = \tan^{-1}(\ln x^2) \\ y' = \frac{\frac{2x}{x^2}}{1 + (\ln x^2)} = \frac{2}{x(1 + (\ln x)^2)}$$

$$\textcircled{4} \quad y = \ln \sqrt{3 \sin 2x} \\ = \frac{1}{2} \ln(3 \sin 2x) \\ \frac{dy}{dx} = \frac{1}{2} \cdot \frac{3 \cos 2x \cdot 2}{3 \sin 2x} = \cot 2x$$

$$\textcircled{5} \quad y = \ln 3\pi + \sin(\ln x^2) \\ y' = 0 + \cos(\ln x^2) \cdot \frac{2x}{x^2}$$

$$\textcircled{6} \quad y = \ln(x^{10} \sin x) = 10 \ln x + \ln \sin x \\ y' = \frac{10}{x} + \cot x$$

$$\textcircled{7} \quad y = \ln \frac{x^2(1+x^2)^4}{\sqrt[3]{1-x^4}}$$

$$\begin{aligned} y &= \ln x^2 \cdot (1+x^2)^4 - \ln \sqrt[3]{1-x^4} \\ &= \ln x^2 + \ln(1+x^2)^4 - \ln \sqrt[3]{1-x^4} \\ &= 2 \ln x + 4 \ln(1+x^2) - \frac{1}{3} \ln(1-x^4) \end{aligned}$$

$$y' = \frac{2}{x} + \frac{4 \cdot 2x}{1+x^2} - \frac{1}{3} \frac{-4x^3}{1-x^4}$$

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$$\textcircled{8} \quad y = \frac{\tan x \cdot e^x \cdot \sqrt{x^2+1}}{\sqrt{x} \cdot \cos x}$$

$$\Rightarrow \ln y = \ln \left[\frac{\tan x \cdot e^x \cdot \sqrt{x^2+1}}{\sqrt{x} \cdot \cos x} \right]$$

$$\Rightarrow \ln y = \ln \tan x + x + \frac{1}{2} \ln(x^2+1) - \frac{1}{2} \ln x - \ln \cos x$$

$$\frac{1}{y} \cdot y' = \frac{\sec^2 x}{\tan x} + 1 + \frac{1}{2} \frac{2x}{x^2+1} - \frac{1}{2} \frac{1}{x} - \frac{-\sin x}{\cos x}$$

$$\begin{aligned} y' &= y \left[\frac{\sec^2 x}{\tan x} + 1 + \frac{x}{x^2+1} - \frac{1}{2x} + \tan x \right] \\ &= \wedge [\end{aligned}$$

⑨ Find $\frac{dy}{dx}$ if $y = \sqrt{\frac{\sin x + \cos x}{\cos x - \sin x}}$

using logarithmic differentiation.

let $\ln y = \ln \sqrt{\frac{\sin x + \cos x}{\cos x - \sin x}}$

$$\Rightarrow \ln y = \frac{1}{2} [\ln(\sin x + \cos x) - \ln(\cos x - \sin x)]$$

$$\frac{1}{y} \cdot y' = \frac{1}{2} \left[\frac{\cos x - \sin x}{\sin x + \cos x} - \frac{-\sin x - \cos x}{\cos x - \sin x} \right]$$

$$y' = \frac{1}{2} y \left[\frac{(\cos x - \sin x)^2 + (\sin x + \cos x)^2}{\cos^2 x - \sin^2 x} \right]$$

$$= \frac{1}{2} y \left[\frac{2}{\cos 2x} \right] = y \sec 2x.$$

Important p 221

1. $\frac{d}{dx} (a^b) = 0$ (24)

2. $\frac{d}{dx} (f(x))^b = b [f(x)]^{b-1} f'(x)$ (x4)

3. $\frac{d}{dx} (a^{g(x)}) = a^{g(x)} \cdot \ln a \cdot g'(x)$ (2^x)

4. What about $\frac{d}{dx} [f(x)]^{g(x)}$? (x^{sin x})

MUST introduce LN!

$$\textcircled{1} \quad \frac{dy}{dx} \quad \text{if} \quad y = x^{(1+\ln x)}$$

$$\text{let} \quad \ln y = (\underline{1+\ln x}) \ln x \quad (\text{product})$$

$$\frac{1}{y} \cdot y' = \frac{1}{x} \ln x + (1+\ln x) \frac{1}{x}$$

$$y' = y \left[\frac{2}{x} \ln x + \frac{1}{x} \right].$$

$$\textcircled{2} \quad \frac{dy}{dx} \quad \text{if} \quad y = (\sec x)^{2x+1}$$

$$\text{let} \quad \ln y = (2x+1) \ln \sec x$$

$$\frac{1}{y} y' = 2 \ln \sec x + (2x+1) \frac{\sec x \tan x}{\sec x}$$

$$\Rightarrow y' = \dots\dots\dots$$

$$\textcircled{3} \quad \frac{dy}{dx} \quad \text{if} \quad y = x^y$$

$$\text{let} \quad \ln y = y \ln x \quad (\text{Implicit!!!})$$

$$\frac{1}{y} y' = y' \ln x + y \cdot \frac{1}{x}$$

$$y' = \dots\dots\dots$$

Worksheet

1. $\frac{d}{dx} \ln(\sin(x^3))$

2. $\frac{d}{dx} \sin(\ln(1-3x))$

3. $\frac{d}{dt} t^4 \ln(1-e^{4t})$

4. $\frac{d}{dx} \frac{6^x \sin 2x}{e^{9x} \tan 4x}$

5. $\frac{d}{dx} (1 - \tan 4x)^5$

6. $\frac{d}{dx} 4^{\sin 3x}$

7. $\frac{d}{dx} (4x)^{\cos 3x}$

8. $\frac{d}{dx} \cos x \cdot x^3$

9. $\frac{d}{dx} (4x)^{\cos x}$

10. $\frac{d}{dx} \frac{2^{3x} \sin(4x)}{\cos(1-x) \operatorname{cosec}(e^{-x})}$